

WHOM TO SCREEN?

INDIVIDUALS WITH FOLLOWING RISK FACTOR/S

TARGET POPULATION : AGE > 40 YRS OF AGE

RISK FACTORS

- Tobacco smoking (pack year ≥ 10)
- Environmental (second hand) tobacco smoke
- Exposure to indoor biomass fuel (fire woods, stoves, coal, cow patty)
- Occupational exposure (dust, fumes)
- Outdoor air pollution (dust, fumes)
- Past history of pulmonary tuberculosis
- Recurrent lower respiratory tract infections
- Untreated and uncontrolled Asthma
- Age of starting smoking

Pack year = Number of cigarettes smoked per day X
Number of years smoked / 20

SCREENING

Suspect COPD if patient has risk factors
with any of the following symptoms:



DIFFICULTY BREATHING WHICH IS

Persistent and progressive
Characteristically worse with exertion
Progressive over time



CHRONIC COUGH WITH SPUTUM

Chronic cough with sputum at least for
3 consecutive months for 2 consecutive
years



NOISY BREATHING OR WHEEZING

DURING ACUTE EXACERBATION PATIENT WILL HAVE:

Increased difficulty in breathing and cough, purulent
sputum, increased sputum volume and more noisy
breathing

Disclaimer: This leaflet is currently being piloted and may contain inconsistencies.
Please help us improve it by reporting any issues to KHDC Nepal.

DIFFERENTIAL DIAGNOSIS: BRONCHIAL ASTHMA

	SUGGEST COPD	SUGGEST BRONCHIAL ASTHMA
Exposure to Smoking/ Biomass	Present	More often absent
Age of onset	After 40 years	Childhood/early adulthood
Nature	Progressive	Intermittent
Initial symptom	Shortness of breath	Cough and wheezing
Diurnal variation	Little	Worse at night and early morning
History of allergies	Absent	Present
Clinical salbutamol response	Partial or no response	Good response
Peak Expiratory Flow Rate (PEFR)	Smaller response makes a diagnosis of COPD more likely	PEF improves by 20% after 2 puffs of salbutamol at 15 minutes
Change in FEV1 (Spirometry)	Smaller response makes a diagnosis of COPD more likely	12% or 200 ml after 2 puff of salbutamol 15 minutes

Other differential diagnosis: Tuberculosis

CHF (Congestive Heart Failure)

Bronchogenic carcinoma



INDICATION FOR LONG TERM OXYGEN THERAPY (REFER)

(AT LEAST 15 HOURS/DAY)

- Two pulse oximetry readings 3 weeks
apart are: $\leq 88\%$ OR $< 89\%$ with Edema
or P pulmonale in ECG

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CHRONIC OBSTRUCTIVE PULMONARY DISEASE (COPD) IDENTIFICATION AND MANAGEMENT FOR PRIMARY CARE PROVIDERS

USE THIS GUIDE TO

- Identify patients with Chronic obstructive
pulmonary disease (COPD)
- Manage and refer patients with COPD

Chronic obstructive pulmonary disease (COPD) is
characterized by persistent respiratory symptoms and
airflow obstruction usually caused by significant exposure
to harmful gases (tobacco smoking and long term
exposure to smoke from biomass fuel)

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DIAGNOSIS

A spirometry is required for a diagnosis of COPD with a FEV1/FVC ratio below 0.7 after bronchodilator. This signifies IRREVERSIBLE OBSTRUCTION

PEAK EXPIRATORY FLOW RATE (PEFR) IS USED TO DIFFERENTIATE BETWEEN COPD AND BRONCHIAL ASTHMA (REFER TO BACK OF THE LEAFLET)

INVESTIGATIONS DURING ACUTE EXACERBATION

- Pulse oximetry,
 - Chest X-ray*
 - Electrocardiography
 - Complete blood count*
- *If available

CLINICAL SEVERITY ASSESSMENT BY MMRC GRADE

MODIFIED MEDICAL RESEARCH COUNCIL (MMRC)	GRADING
Dyspnea only with strenuous exercise	0
Dyspnea on hurrying or walking slight uphill	1
Walks slower than people of same age or have to stop to breathe while walking at their own pace	2
Stop to breathe on walking 100 meters or after a few minutes	3
Too dyspneic to leave house or dyspnea when dressing	4

Ref: Global Initiative for Obstructive Lung Disease (GOLD) 2024



NON-PHARMACOLOGICAL APPROACH

Smoking cessation: Refer to KHDC Smoking Cessation Leaflet

- Reduce environmental dust by using damp cloths to clean furniture
- Avoid cooking using firewood, stove or any other biomass producing like cow patty
- Well ventilated rooms
- Wash hands/ Use face masks
- Vaccination*: Tdap, Pertussis, Influenza, pneumococcal, COVID-19, (H.Zooster, RSV * if available)
- Physical exercise & yoga
- Consumption of nutritious food (Carbohydrate and calcium rich)



PHARMACOLOGICAL TREATMENT

SEVERITY

≥ 2 moderate exacerbations or ≥ 1 leading to hospitalization

0 or 1 moderate exacerbations (not leading to hospital admission)

TREATMENT

LABA + LAMA* consider LABA+LAMA+ICS* if blood eosinophil ≥ 300

LABA or LAMA

mMRC 0-1

(LAMA) + (LABA)

MRC ≥ 2

LABA: Long-Acting Beta-Agonists
LAMA: Long-Acting Muscarinic Antagonist
ICS: Inhaled Corticosteroid.



INITIAL MANAGEMENT OF ACUTE EXACERBATION

Tab. Prednisolone 40 mg stat and after breakfast for 5 days
Tab. Co-amoxiclav 625 mg po TDS for 5 days, and
Tab Azithromycin 500 mg po od for 3 days
Salbutamol MDI with Spacer 2 puffs every 20 min for 1 hours then every 4 hourly until recovery
Nebulisation Ipratropium/Salbutamol hourly 2-3 hours
Low flow 2 litres/min oxygen by nasal prong
Oxygen target 90%

REFER

CRITERIA FOR REFERRAL

- Acute exacerbation after initial management
- Uncontrolled symptoms inspite of LAMA+LABA
- Onset of symptoms under 40 years
- Frequent chest infections (to exclude bronchiectasis)
- Onset of right heart failure (swelling of the legs)
- Hemoptysis (exclude carcinoma of the lung)
- Fever and Weight loss
- If the patient is not getting better after initial management
- In stable COPD if SPO2 is <88% , referral for Long Term Oxygen Therapy Assessment

FOLLOW-UP

Initial follow-up should be done after one month then after 3-6 months if stable



SMOKING CESSATION IS OF UTMOST IMPORTANCE



CONTACT US

For any clinical queries / If anything in the document is difficult to understand



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